Roll No. ....

11471

B. Pharma 1st Sem.
(Regular/Re-appear/Improvement)
(Mercy Chance)
Examination – December, 2023

HUMAN ANATOMY AND PHYSIOLOGY II-THEORY

Paper : BP101T

Time : Three hours] *[Maximum Marks : 75*

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-**A** is *compulsory*, each carry **2** marks. Attempt any *seven* questions out of nine from Section-**B**, each carry **5** marks and *two* questions out of three from Section-**C**, each carry **10** marks.

SECTION – A

Note : Attempt *all* questions : $2 \times 10 = 20$

1. (a) Define Homeostasis.

(b) Define Cytokinesis.

(c) What is Bone remodeling ?

(d) What is the function of Synovial fluid ?

(e) What are Lymphatic ducts ?

(f) What is Leukopoiesis ?

(g) Define Neurotransmission.

(h) Enlist the functions of Spinal cord.

(i) Write a note on Endocarditis.

(j) Write the sign and symptoms of coronary artery disease.

SECTION – B

Note : Attempt any *seven* questions : $5 \times 7 = 35$

2. Describe the structure and functions of Endoplasmic reticulum.

3. Explain the interphase stage in somatic cell division.

4. Describe different types of connective tissues.

5. Write a detailed note on histology of bone tissue.

6. Write down the structural and functional classifications of Joints.

7. What is Neuromuscular junction ? Describe its significance.

8. Define hemopoeisis. Describe the significance of Reticulo endothelial system.

9. Discuss the functions of lymphatic system.

10. Discuss the various factors necessary for Erythropoiesis.

SECTION – C

Note : Attempt any *two* questions : $2 \times 10 = 20$

11. Describe the characteristics features of sympathetic and parasympathetic nervous system.

12. What are Spinal nerves ? Write a note on feedback regulation in Erythropoiesis process.

13. Define Atrial systole. Explain different phases of action potential generation in cardiac muscle.

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 1st Semester, 1st Sessional Examination, Feb. 2023
Paper Code –BP101T; Human Anatomy and Physiology-I (Theory)

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Duration: 1 hr 30 minutes **Total marks: 30**

1. **Question no. 1 is compulsory.** **(1×1=10 marks)**

i) Myology is the study of:
   a. Bones
   b. Muscle
   c. cartilage
   d. all the above

ii) Haemopoiesis is a process of the production of
   a. Blood Plasma
   b. Erythrocytes
   c. Bone Marrow
   d. Haemoglobin

iii) Which of the following is not required for clot formation
   a. Vitamin K
   b. Calcium
   c. Plasmin
   d. Fibrinogen

iv) Haemopoiesis is a process of the production of
   a. Blood Plasma
   b. Erythrocytes
   c. Bone Marrow
   d. Haemoglobin

v) The function of protein sorting, lipid sorting, and secretion is performed by
   a. Rough ER
   b. Smooth ER
   c. Golgi apparatus
   d. Peroxisomes

vi) Discovery of blood group was done by
   a. Karl Landsteiner
   b. Paul Ehrlich
   c. Ogston

vii) The urinary bladder is located in the region.........
   a. Umbilical
   b. Hypogastric
   c. Right Inguinal
   d. Left Inguinal

viii) The term "suicidal bags" is used for
   a. Lysosomes
   b. Peroxisomes
   c. Mitochondria
   d. Ribosomes

ix) The study of blood vessels is called
   a. Cardiology
   b. Haematology
   c. Angiology
   d. Serology

x) Chromosome duplication occurs in which of the following phases?
   a. G1 Phase
   b. G2 Phase
   c. S Phase
   d. Prophase

2. **Attempt any two questions from the rest.** **(2×5=10 marks)**
   a. What is haemopoiesis? Describe the process of haemopoiesis.
   b. Write a note on homeostasis and its regulation.
   c. Discuss the structure of heart with a neat, labelled diagram.

3. **Attempt anyone questions from the rest.** **(1×10=10 marks)**
   a. What is blood pressure? Give normal value of systolic and diastolic blood pressure and explain regulation of blood pressure.
   b. Define Tissue and classify them? Write down and three difference between muscular and nervous tissue.

B. Pharmacy 1st Semester 2nd Sessional Exam April 2023
Human Anatomy and Physiology-I (BP104T)

Time: 1.5 Hrs Max. Marks: 30

SECTION-A

Attempt all Questions. Each question carry 2 marks. **2X5=10**

1. Write the function of Eye.
2. Write the division of skeletal system
3. Classification of peripheral nervous system
4. Function of appendicular skeletal system.
5. Functions of lymphatic system .

SECTION-B

Short Answer Questions. Attempt any 2 out of Three. **5X2=10**

4. Explain the functions of sympathetic and parasympathetic nervous system.
5. Describe the Structure and functions of skin.
6. What are cranial nerves? Explain the different types of cranial nerves in detail.

SECTION-C

Long Answer type Question. Attempt any One out of Two **10X1=10**

3. Describe physiology of muscle contraction with example.
4. Explain different types of joints, movements and its articulation.

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy 1st Semester; Second Sessional Examination

Pharmaceutical Inorganic Chemistry (Paper code –BP 104T)

Date- March 31, 2023

Duration- 1h 30 min Max. Marks- 30

Question no.1 is compulsory. Each question carry 2 marks.

1. **Answer the following.**
   a. Haematinics with one examples
   b. Expectorants with one examples
   c. What is radioactivity. Give its units
   d. What are radioisotopes. Give its examples.
   e. Define buffer. Write the equation and its name used for calculating pH of buffer solution.

$CuSO_4$, $FeSO_4$

Attempt any two questions. Each question carry 5 marks. AgCl , U_{238}

2. Give method of preparation of ammonium chloride, copper sulphate and ferrous sulphate?
3. Write properties of alpha, beta and gamma radiations. Write a note on I^{131} as radioisotope.
4. What are saline cathartics? Write the preparation and assay of magnesium sulphate.

Attempt any one question. Each question carry 10 marks.

5. Molecular formula, properties, method of preparation, name of test for identification and uses of Potassium iodide and Sodium nitrite.
6. Discuss about measurement of radioactivity. Give any one method in detail. $CuSO_4$

NH_4Cl. NH_4Cl

Rd.

$H_2O + NH_4 + NaCl_2 \xrightarrow{Heat} NH_4Cl + HCl + NaCl$

HNO_3

$NaNO_3$

Department of Pharmaceutical Sciences,

B. Pharmacy 1st Semester-2023 2nd Sessional

BP-102T P'ceutical Analysis

Time: 1.5 Hours | Marks: 30

Section-A

Attempt all questions. Each question carries 2 marks. **2x5=10**

Q.1 Define the following:

a) Define equivalent conductance.

b) Define co-precipitation and post-precipitation.

c) What is Nernst equation?

d) Give applications of conductometry?

e) What are indicator electrodes explain with example?

Section-B

Long Answer Questions: Answer any two out of the following three questions **10x1=10**

Q.2 Write a detailed note on silver-silver chloride electrode and SHE with diagram.

Q.3 Write an illustrative note on diazotization titration.

Section-C

Short Answer Questions: Answer any seven out of the following nine questions **5x2=10**

Q.5 Write a note on steps involved in gravimetric analysis.

Q.6 Discuss iodometry and iodimetry in details.

Q.13 Give a note on Dropping Mercury electrode and Rotating Platinum electrode.

Department of Pharmaceutical Sciences,
Indira Gandhi University, Meerpur, Rewari (Haryana)

B. Pharmacy 1st Semester, 2nd Sessional Theory, 29-March-2023, BP-103T Pharmaceutics-I

Time: 1.5 Hours Marks: 30

Section-A

Attempt all questions. Each question carries 2 marks. 2x5=10

Q.1 Define the following:

a) Define therapeutic incompatibility with example. What are the different types of incompatibilities?

b) Write down different water soluble & water miscible bases used in the preparation of suppository.

c) Define geometric dilution of powders.

d) Gibb's free energy & its significance

e) Define the term 'powder'.

Section-B

Long Answer Questions: Answer any two out of the following three questions 10x1=10

Q.2 Define therapeutic incompatibility. What are the different types of incompatibilities? Discuss in detail about chemical incompatibilities.

Q.3 Classify Emulsion & write down their identification tests.

Section-C

Short Answer Questions: Answer any seven out of the following nine questions 5x2=10

Q.5 Write different method of preparation of suppository. Explain hot or fusion method of preparation.

Q.6 Define the term 'powder'. Classify different types of powders. Discuss the bulk powders which are meant for external use.

Q.7 Discuss the various sources of errors while dispensing a prescription. How are these errors corrected?

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy, Ist semester Final Examination

BP 106RMT: Remedial Mathematics, (QP ID: 11477)

Reflection Scalar All non-zero Repeating Multiples

Duration- 1h 30 min **Max. Marks- 35**

Attempt any one out of two. Each question carries 10 marks.

(10) **Q1.** Explain any five properties of determinants with examples.

Q2. Find the equation of the line perpendicular to the line $x-7y+5=0$ and having x-intercept 3.

Attempt any five out of seven. Each question carries 5 marks.

Q3. Differentiate between singular and non-singular matrices with the help of an example.

Q4. Differentiate w.r.t. x: $2x - \frac{3}{x^2} + 4\sqrt[3]{x} + 2\sin x$.

Q5. Write application of Partial Fraction in Chemical Kinetics and Pharmacokinetics.

Q6. Define order and degree of a differential equation by giving suitable examples.

Q7. Find the solution for the differential equation $(2xy-\sin x)dx + (x^2-\cos y)dy = 0$.

(2) $x^{(2)}$

Q8. On what intervals is $f(x) = \frac{2}{x-3} + 3$ continuous?

Q9. Write some properties of logarithms with examples.

(20) $\frac{dy}{dx} = \frac{(2xy - \sin x)}{(x^2 - \cos)}$

(30)

$2x - \frac{3}{x^2} + 4^{-}x + 2\sin^3 x$

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Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 1st Semester, Subject: Remedial Maths, 2nd Sessional Examination
Paper Code: BP104T, March-2023

Duration: 1hr **Total marks: 30**

Q 1. Attempt any four out of six: **(4 X 5=20 marks)**

a) For the function $f(x) = x^2 + 2x + 4$, find $f'(x)$ by first principle.

b) Differentiate the function $y = \frac{1}{8}x^8 + \frac{1}{5}x^5 - \frac{4}{3}x^3 + 2\ w.r.t.x$

3 c) Differentiate the function $\frac{x^3 + e^x}{logx+5}\ w.r.t.x$

d) Find the coordinates of the incenter of the triangle whose vertices are $(4,-2), (-2,4) \& (5,5)$

e) Prove that angle between the lines $27x - 18y + 25 = 0 \ \& \ 2x + 3y + 7 = 0\ is\ 90°$

2 f) Prove that $3\,log4 + 2\,log5 + \frac{1}{3}log64 - \frac{1}{2}log16 = 2$

Q 2. Attempt any one out of two: **(1 X 10=10 marks)**

a) Find $\frac{d^3y}{dx^3}$ if $y = 2sin2x + cosx$

b) Solve the following.

(i) Find the equation of ~~straight line which makes an angle~~ of 135° with the x-axis and passes through the point (3,3).

1 (ii) Determine the angle B, if A (-2,1), B (2,3) and C (-2, -4) are the vertices of triangle.

2x

3x5x+

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences

B. Pharm. 1st Semester, Final Theory Examination,27 Jan 2026

Paper Code -- BP106RMT Remedial Mathematics

Total marks: 35

Duration: 1.30 hrs

Attempt one out of two questions (10 ×1 = 10)

Q 1. Find the Adjoint of $\begin{bmatrix} 1 & 4 & -2 \\ -2 & -5 & 4 \\ 1 & -2 & 1 \end{bmatrix}$.

$a(\ )+b($

$a+b$

Q 2. a) Explain Proper and improper fraction with example. 0

b) Evaluate $\int_0^1 x^2$.ex with definite Integral.

Attempt any five out of seven questions:(5× 5 = 25)

Q 3. Find the inverse of A= $\begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$ using Cayley-Hamilton theorem. 3

Q 4. Explain classification of real valued function with example.

25 Q 5. Find $\frac{d^2y}{dx^2}$. y=sinx+x^4 through Successive differentiation.

Q 6. Application of logarithm to solve pharmaceutical problems. 5

5 Q 7. Show that the points A (2,-4) B (4,-2) C (7,1) are collinear using distance formula.

4 Q 8. Define Order and degree with examples.

Q 9. Evaluation $\int x^2$.cosx dx by Integration by parts.